

Machine Learning Fundamentals

What is Machine Learning?

Machine Learning is a branch of AI that enables computers to learn from data, improve through experience, and make predictions or decisions automatically.

How Machine Learning Works

1. **Collect Data** - Gather historical data.
2. **Train a Model** - Use algorithms to learn patterns.
3. **Test the Model** - Verify accuracy using new data.
4. **Make Predictions** - Use the trained model on real-world data.

Types of Machine Learning

1. Supervised Learning
2. Unsupervised Learning
3. Reinforcement Learning

1. Supervised Learning

The model learns from labeled data (known answers).

- Example: Predicting house prices or email spam detection.
- Algorithms: Linear Regression, Logistic Regression, Decision Trees.

2. Unsupervised Learning

The model discovers hidden patterns in data without labels.

- Example: Customer segmentation.
- Algorithms: K-Means Clustering, Hierarchical Clustering.

3. Reinforcement Learning

The model learns through rewards and penalties.

- Example: Self-driving cars, game-playing AI.

Real-World Applications

- Netflix and YouTube recommendations
- Fraud detection in banking
- Speech recognition (Siri, Alexa)
- Medical diagnosis
- Predictive maintenance
- Autonomous vehicles

AI vs Machine Learning vs Deep Learning

Artificial Intelligence (AI)

└─ Machine Learning (ML)

└─ Deep Learning (DL)

- **AI:** Makes machines behave intelligently.
- **ML:** Enables systems to learn from data.
- **Deep Learning:** Uses neural networks to solve complex problems such as image and speech recognition.

Example

Suppose you provide a model with historical employee attrition data:

- Age
- Experience
- Salary
- Department

The ML model learns patterns and predicts whether a current employee is likely to leave the organization.

ML Terminologies

1. Dataset

A collection of data used for training and testing machine learning models.

Example: Employee data containing Age, Experience, Salary, and Attrition Status.

2. Feature

An input variable used to make predictions.

Example:

- Age
 - Salary
 - Experience
-

3. Label (Target Variable)

The output that the model predicts.

Example:

- Employee Left = Yes/No
 - House Price = ₹50 Lakhs
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4. Training Data

The data used to teach the machine learning model.

5. Testing Data

New data used to evaluate how well the model performs.

6. Model

A mathematical representation that learns patterns from data and makes predictions.

Examples:

- Linear Regression
 - Decision Tree
 - Random Forest
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7. Algorithm

The technique used to build the model.

Examples:

- Logistic Regression
 - K-Means Clustering
 - Support Vector Machine (SVM)
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8. Training

The process of teaching the model using historical data.

9. Prediction

The output generated by a trained model.

Example: Predicting whether a customer will purchase a product.

10. Accuracy

Measures how many predictions were correct.

Accuracy = Correct Predictions / Total Predictions

11. Overfitting

The model memorizes training data and performs poorly on new data.

12. Underfitting

The model fails to learn enough patterns from the data.

13. Feature Engineering

Creating new useful features from existing data.

Example:

Date of Birth → Age

This is a key concept in traditional machine learning.

14. Hyperparameters

Settings configured before training.

Examples:

- Number of Trees
 - Learning Rate
 - Epochs
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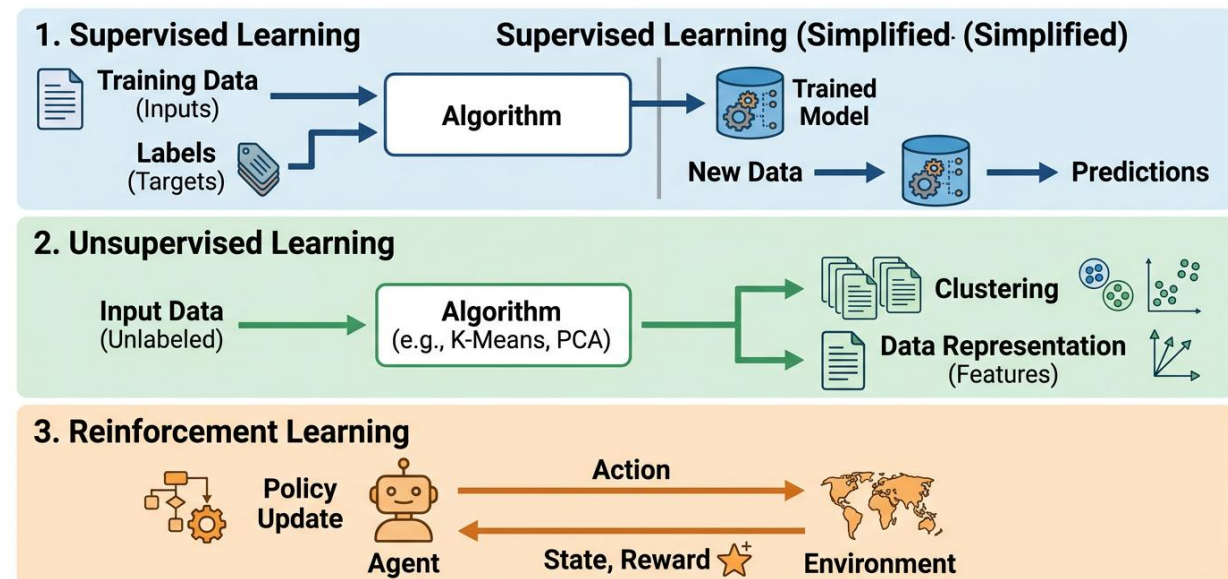
15. Confusion Matrix

Used to evaluate classification models.

Actual	Predicted
True Positive	Correct Positive
True Negative	Correct Negative
False Positive	Wrong Positive
False Negative	Wrong Negative

Types of Machine Learning

Machine Learning Paradigms



1. Supervised Learning

The model learns from **labeled data**.

Input + Expected Output are provided during training.

Examples

- Email Spam Detection
- House Price Prediction
- Employee Attrition Prediction

Algorithms

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forest
- SVM

Diagram

Input Data



Train Model



Known Output



Prediction

2. Unsupervised Learning

The model learns from **unlabeled data**.

It identifies hidden patterns and groups.

Examples

- Customer Segmentation
- Product Recommendation Groups
- Market Basket Analysis

Algorithms

- K-Means Clustering
- Hierarchical Clustering
- PCA

Diagram

Input Data

↓

Find Patterns

↓

Groups / Clusters

3. Reinforcement Learning

The model learns through **trial and error**.

It receives rewards or penalties based on actions.

Examples

- Self-Driving Cars
- Chess Playing AI
- Robotics

Diagram

Agent



Action



Environment

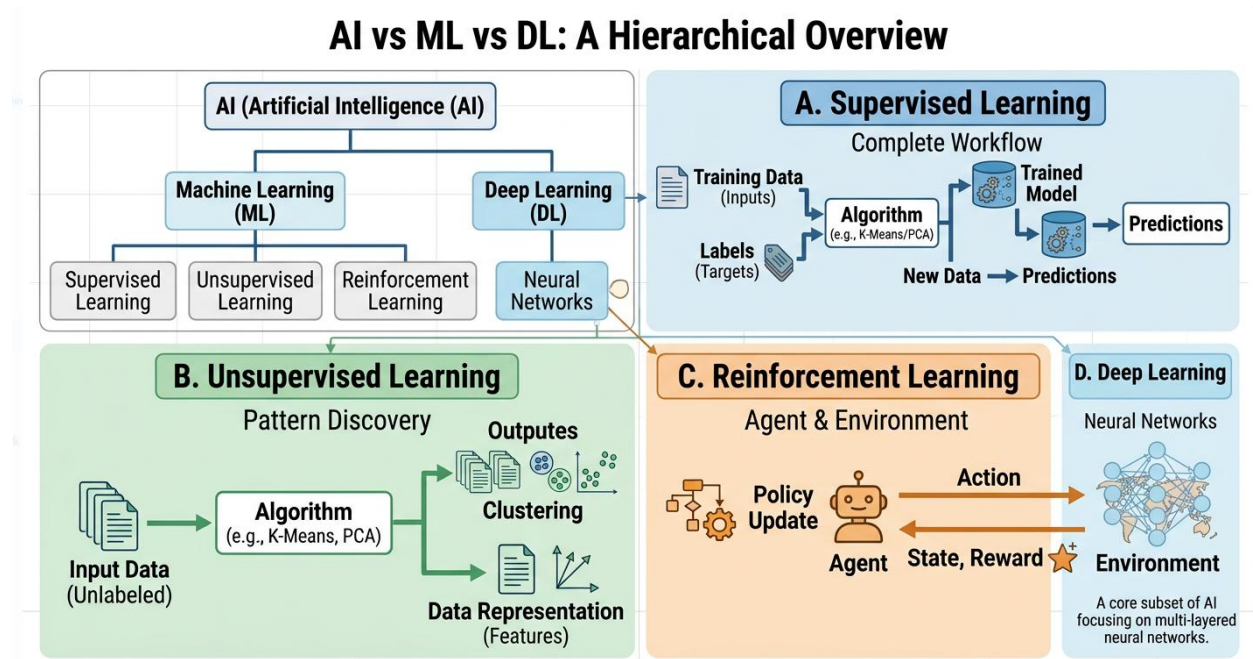


Reward / Penalty



Learning

AI vs ML vs DL



Deep Learning automates feature extraction using multiple neural network layers and is commonly used for images, speech, and Generative AI applications.

Training and Testing Data

Machine Learning models learn from historical data and are then evaluated using unseen data.

Training Data

Training data is the dataset used to teach the model and help it learn patterns and relationships.

Example

Age	Experience	Salary	Left Company
25	2	5 LPA	Yes
35	8	12 LPA	No
40	12	15 LPA	No

The model learns patterns from these records.

Testing Data

Testing data is a separate dataset used to evaluate how well the trained model performs on new, unseen data.

Example

Age	Experience	Salary
30	5	8 LPA

The model predicts:

Left Company = No

Why Split the Data?

If we test using the same data used for training, the model may simply memorize the answers.

The split helps measure how well the model generalizes to new data.

Typical split:

Training Data = 80%

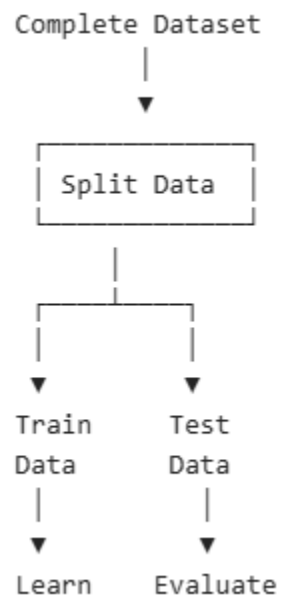
Testing Data = 20%

Or

Training Data = 70%

Testing Data = 30%

Visual Representation



Machine Learning Lifecycle

The ML lifecycle is the end-to-end process of building, evaluating, deploying, and maintaining machine learning solutions. Organizational learning content references collecting, preprocessing, training, evaluating, and deploying models as part of a complete ML project.

Step 1: Problem Definition

Identify the business problem.

Examples

- Predict employee attrition
 - Detect fraud transactions
 - Forecast sales
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Step 2: Data Collection

Gather data from sources such as:

- SQL Server
 - CSV files
 - APIs
 - Azure Data Lake
 - Business applications
-

Step 3: Data Preprocessing

Prepare the data before training.

Activities include:

- Handling missing values
- Removing duplicates
- Data cleaning
- Feature scaling
- Encoding categorical values

Step 4: Feature Engineering

Create meaningful features that improve model performance.

Example:

Date Of Birth

↓

Age

Step 5: Train the Model

Choose an algorithm and train it.

Examples:

- Linear Regression
 - Logistic Regression
 - Decision Tree
 - Random Forest
 - SVM
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Step 6: Model Evaluation

Measure model performance using:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

The ML curriculum includes these evaluation concepts.

Step 7: Model Deployment

Deploy the trained model to production environments.

Examples:

- Azure Machine Learning
- Flask APIs
- Web Applications

Model deployment is part of the end-to-end ML learning path.

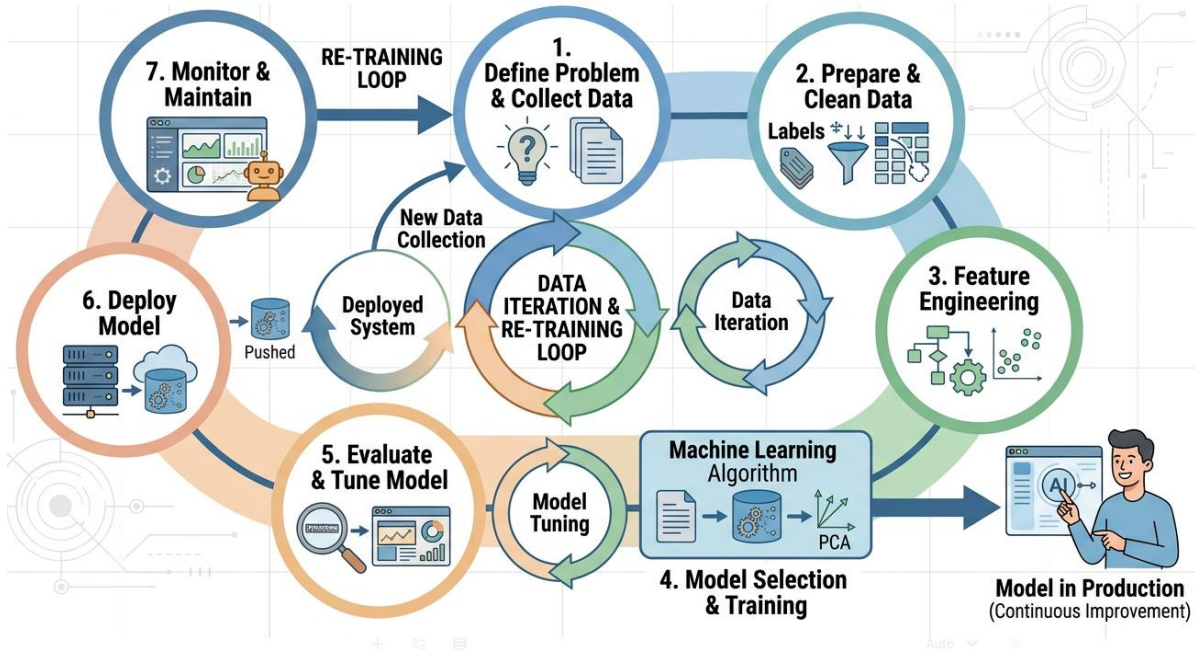
Step 8: Monitoring and Maintenance

Monitor production performance and retrain models when needed.

Enterprise ML lifecycle content highlights monitoring, evaluation, retraining, deployment, and operationalization as ongoing activities.

Complete Lifecycle Diagram

MACHINE LEARNING LIFE CYCLE



REAL-WORLD AZURE EXAMPLE: EMPLOYEE ATTRITION PREDICTION WORKFLOW

